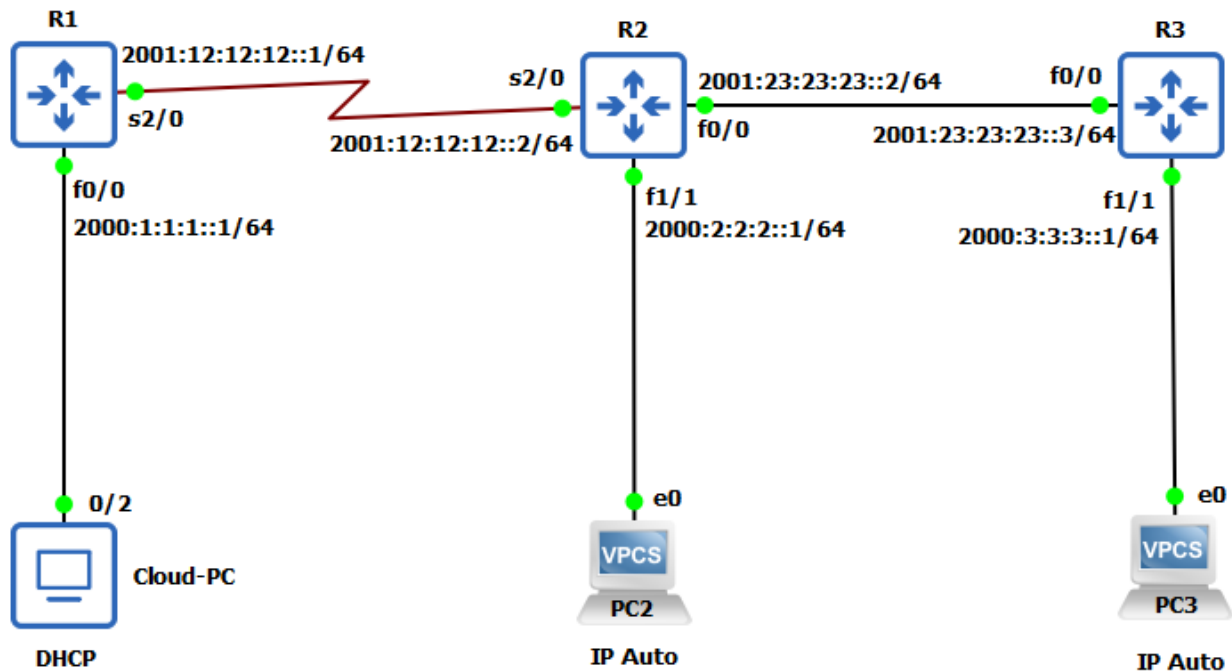


3 Routers Initial Configuration – IPv6



Task

1. Configure routers with IPv6 address as shown in topology
2. Configure enable password as **ccna** and device access password as **cisco**
3. Configure 16 remote access sessions for routers.
4. Remote session should log off after 1 min of idle timeout.
5. Configure PC with IP address and default gateway.

Note: Cloud-PC is a cloud which bridge the R1 router with loopback adapter on real PC.

R1 Router Configuration:

```

Router>en
Router#enable
Router#config t
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#ipv6 unicast-routing
R1(config)#int fa0/0
R1(config-if)#ipv6 address 2000:1:1:1::1/64
R1(config-if)#description Link to local LAN
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int s2/0
R1(config-if)#ipv6 address 2001:12:12:12::1/64

```

```

R1(config-if)#description Link to R2 on s2/0
R1(config-if)#encapsulation hdlc
R1(config-if)#clockrate 64000
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#line console 0
R1(config-line)#password cisco1
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#exit
R1#

```

R2 Router Configuration:

```

Router#
Router#enable
Router#config t
Router(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#ipv6 unicast-routing
R2(config)#int fa1/1
R2(config-if)#ipv6 address 2000:2:2:2::1/64
R2(config-if)#description Link to local LAN
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int s2/0
R2(config-if)#ipv6 address 2001:12:12:12::2/64
R2(config-if)#description Link to R1 on s2/0
R2(config-if)#encapsulation hdlc
R2(config-if)#clockrate 64000
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa0/0
R2(config-if)#ipv6 address 2001:23:23:23::2/64
R2(config-if)#description Link to R3 on fa0/0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 1 0

```

```
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password cisco1
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#exit
R2#
```

R3 Router Configuration:

```
Router>en
Router#enable
Router#config t
Router(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable password ccna
R3(config)#ipv6 unicast-routing
R3(config)#int fa1/1
R3(config-if)#ipv6 address 2000:3:3:3::1/64
R3(config-if)#description Link to local LAN
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int fa0/0
R3(config-if)#ipv6 address 2001:23:23:23::3/64
R3(config-if)#description Link to R2 on fa0/0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#line vty 0 15
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 1 0
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password cisco1
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#exit
R3#
```

PC2 Configuration:

```
PC2> ip auto
GLOBAL SCOPE : 2000:2:2:2:2050:79ff:fe66:6801/64
ROUTER LINK-LAYER : ca:02:0c:24:00:1d
```

PC3 Configuration:

PC3> ip auto

GLOBAL SCOPE : 2000:3:3:3:2050:79ff:fe66:6800/64

ROUTER LINK-LAYER : ca:03:1e:00:00:1d

Cloud-PC

Ethernet adapter Loopback:

```

Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2000:1:1:1:d5d6:56d2:75d0:b647
Temporary IPv6 Address. . . . . : 2000:1:1:1:4de9:2bba:4e97:eca9
Link-local IPv6 Address . . . . . : fe80::d5d6:56d2:75d0:b647%24
IPv4 Address. . . . . : 192.168.1.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::c801:20ff:fe88:8%24
                           192.168.1.1

```

✓ Verification & Testing:**R1#sh ipv6 int brief**

FastEthernet0/0 [up/up]

FE80::C801:20FF:FE88:8

2000:1:1:1::1

FastEthernet0/1 [administratively down/down]

unassigned

FastEthernet1/0 [administratively down/down]

unassigned

FastEthernet1/1 [administratively down/down]

unassigned

Serial2/0 [up/up]

FE80::C801:20FF:FE88:8

2001:12:12:12::1

Serial2/1 [administratively down/down]

unassigned

Serial2/2 [administratively down/down]

unassigned

Serial2/3 [administratively down/down]

unassigned

R2#sh ipv6 int brief | exclude unassigned

FastEthernet0/0 [up/up]

FE80::C802:CFF:FE24:8

2001:23:23:23::2

FastEthernet0/1 [administratively down/down]

FastEthernet1/0 [administratively down/down]

FastEthernet1/1 [up/up]

FE80::C802:CFF:FE24:1D

2000:2:2:2::1

Serial2/0 [up/up]

FE80::C802:CFF:FE24:8

2001:12:12:12::2

```
Serial2/1      [administratively down/down]
Serial2/2      [administratively down/down]
Serial2/3      [administratively down/down]
```

R3#sh ipv6 int brief | exclude unassigned/down

```
FastEthernet0/0  [up/up]
  FE80::C803:1EFF:FE00:8
  2001:23:23:23::3
FastEthernet1/1  [up/up]
  FE80::C803:1EFF:FE00:1D
  2000:3:3:3::1
```

R1#sh int description | exclude down

Interface	Status	Protocol	Description
Fa0/0	up	up	Link to local LAN
Se2/0	up	up	Link to R2 on s2/0

R2#sh int description | exclude down

Interface	Status	Protocol	Description
Fa0/0	up	up	Link to R3 on fa0/0
Fa1/1	up	up	Link to local LAN
Se2/0	up	up	Link to R1 on s2/0

R3#sh int description | exclude down

Interface	Status	Protocol	Description
Fa0/0	up	up	Link to R2 on fa0/0
Fa1/1	up	up	Link to local LAN

R1#sh run | section interface

```
interface FastEthernet0/0
description Link to local LAN
no ip address
duplex auto
speed auto
ipv6 address 2000:1:1:1::1/64
interface FastEthernet0/1
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/0
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/1
no ip address
shutdown
duplex auto
speed auto
interface Serial2/0
```

```
description Link to R2 on s2/0
no ip address
ipv6 address 2001:12:12:12::1/64
serial restart-delay 0
clock rate 64000
interface Serial2/1
no ip address
shutdown
serial restart-delay 0
interface Serial2/2
no ip address
shutdown
serial restart-delay 0
interface Serial2/3
no ip address
shutdown
serial restart-delay 0
```

R2#sh run | section interface

```
interface FastEthernet0/0
description Link to R3 on fa0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:23:23:23::2/64
interface FastEthernet0/1
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/0
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/1
description Link to local LAN
no ip address
duplex auto
speed auto
ipv6 address 2000:2:2:2::1/64
interface Serial2/0
description Link to R1 on s2/0
no ip address
ipv6 address 2001:12:12:12::2/64
serial restart-delay 0
clock rate 64000
interface Serial2/1
no ip address
shutdown
serial restart-delay 0
```

```
interface Serial2/2
no ip address
shutdown
serial restart-delay 0
interface Serial2/3
no ip address
shutdown
serial restart-delay 0
```

R3#sh run | section interface

```
interface FastEthernet0/0
description Link to R2 on fa0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:23:23:23::3/64
interface FastEthernet0/1
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/0
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/1
description Link to local LAN
no ip address
duplex auto
speed auto
ipv6 address 2000:3:3:3::1/64
interface Serial2/0
no ip address
shutdown
serial restart-delay 0
interface Serial2/1
no ip address
shutdown
serial restart-delay 0
interface Serial2/2
no ip address
shutdown
serial restart-delay 0
interface Serial2/3
no ip address
shutdown
serial restart-delay 0
```

R1#sh ipv6 interface fa0/0

FastEthernet0/0 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::C801:20FF:FE88:8

No Virtual link-local address(es):

Description: Link to local LAN

Global unicast address(es):

2000:1:1:1::1, subnet is 2000:1:1:1::/64

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF00:1

FF02::1:FF88:8

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 30000)

ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)

ND router advertisements are sent every 200 seconds

ND router advertisements live for 1800 seconds

ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

R1#sh ipv6 interface s2/0

Serial2/0 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::C801:20FF:FE88:8

No Virtual link-local address(es):

Description: Link to R2 on s2/0

Global unicast address(es):

2001:12:12:12::1, subnet is 2001:12:12:12::/64

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF00:1

FF02::1:FF88:8

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 30000)

ND RAs are suppressed (periodic)

Hosts use stateless autoconfig for addresses.

R2#sh ipv6 interface fa0/0

FastEthernet0/0 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::C802:CFF:FE24:8

No Virtual link-local address(es):

Description: Link to R3 on fa0/0

Global unicast address(es):

2001:23:23:23::2, subnet is 2001:23:23:23::/64

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF00:2

FF02::1:FF24:8

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 30000)

ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)

ND router advertisements are sent every 200 seconds

ND router advertisements live for 1800 seconds

ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

R2#sh ipv6 interface fa1/1

FastEthernet1/1 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::C802:CFF:FE24:1D

No Virtual link-local address(es):

Description: Link to local LAN

Global unicast address(es):

2000:2:2:2::1, subnet is 2000:2:2:2::/64

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF00:1

FF02::1:FF24:1D

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 30000)

ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)

ND router advertisements are sent every 200 seconds

ND router advertisements live for 1800 seconds

ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

R3#sh ipv6 interface fa0/0

FastEthernet0/0 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::C803:1EFF:FE00:8

No Virtual link-local address(es):

Description: Link to R2 on fa0/0

Global unicast address(es):

2001:23:23:23::3, subnet is 2001:23:23:23::/64

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF00:3

FF02::1:FF00:8

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 30000)

ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)

ND router advertisements are sent every 200 seconds

ND router advertisements live for 1800 seconds

ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

R3#sh ipv6 interface fa1/1

FastEthernet1/1 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::C803:1EFF:FE00:1D

No Virtual link-local address(es):

Description: Link to local LAN

Global unicast address(es):

2000:3:3:3::1, subnet is 2000:3:3:3::/64

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF00:1

FF02::1:FF00:1D

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 30000)

ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)

ND router advertisements are sent every 200 seconds

ND router advertisements live for 1800 seconds

ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

Ping the R2 router from PC-2

PC2> ping 2000:2:2:2::1

2000:2:2:2::1 icmp6_seq=1 ttl=64 time=16.213 ms

2000:2:2:2::1 icmp6_seq=2 ttl=64 time=10.076 ms

2000:2:2:2::1 icmp6_seq=3 ttl=64 time=9.317 ms

2000:2:2:2::1 icmp6_seq=4 ttl=64 time=10.124 ms

2000:2:2:2::1 icmp6_seq=5 ttl=64 time=8.251 ms

Ping the R3 router from PC-3

```
PC3> ping 2000:3:3:3::1
2000:3:3:3::1 icmp6_seq=1 ttl=64 time=19.241 ms
2000:3:3:3::1 icmp6_seq=2 ttl=64 time=10.073 ms
2000:3:3:3::1 icmp6_seq=3 ttl=64 time=10.129 ms
2000:3:3:3::1 icmp6_seq=4 ttl=64 time=9.147 ms
2000:3:3:3::1 icmp6_seq=5 ttl=64 time=10.111 ms
```

Ping the R1 router from Cloud-PC

```
C:\Users\Ratan>ping 2000:1:1:1::1

Pinging 2000:1:1:1::1 with 32 bytes of data:
Reply from 2000:1:1:1::1: time=55ms
Reply from 2000:1:1:1::1: time=11ms
Reply from 2000:1:1:1::1: time=6ms
Reply from 2000:1:1:1::1: time=11ms

Ping statistics for 2000:1:1:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 55ms, Average = 20ms

C:\Users\Ratan>
```

Ping the Cloud-PC from R1 router

R1#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 32/48/76 ms

R1#

Ping the PC-2 from R2 router

R2#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/49/96 ms

R2#

Ping the PC-3 from R3 router

R3# ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/50/76 ms

R3#

Telnet R1 router from Cloud-PC

```

Telnet 2000:1:1:1:1

User Access Verification

Password:
% Password: timeout expired!
Password:
R1>en
Password:
R1#sh users
      Line        User           Host(s)          Idle           Location
  *  0 con 0
  *  2 vty 0
                                     idle
                                     00:04:11
                                     00:00:00
                                     2000:1:1:1:4DE9:2BBA:4E97:ECA9

Interface      User           Mode           Idle           Peer Address
R1#

```

R1#sh cdp neighbors detail**Device ID: R2**

Entry address(es):

IPv6 address: 2001:12:12:12::2 (global unicast)

IPv6 address: FE80::C802:CFF:FE24:8 (link-local)

Platform: Cisco 7206VXR, Capabilities: Router

Interface: Serial2/0, Port ID (outgoing port): Serial2/0

Holdtime : 174 sec

Version :

Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M5, RELEASE SOFTWARE (fc2)

Technical Support: <http://www.cisco.com/techsupport>

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advertisement version: 2

R2# sh cdp neighbors detail**Device ID: R3**

Entry address(es):

IPv6 address: 2001:23:23:23::3 (global unicast)

IPv6 address: FE80::C803:1EFF:FE00:8 (link-local)

Platform: Cisco 7206VXR, Capabilities: Router

Interface: FastEthernet0/0, Port ID (outgoing port): FastEthernet0/0

Holdtime : 152 sec

Version :

Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M5, RELEASE SOFTWARE (fc2)

Technical Support: <http://www.cisco.com/techsupport>

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advertisement version: 2

Duplex: full

Device ID: R1

Entry address(es):

IPv6 address: 2001:12:12:12::1 (global unicast)

IPv6 address: FE80::C801:20FF:FE88:8 (link-local)

Platform: Cisco 7206VXR, Capabilities: Router

Interface: Serial2/0, Port ID (outgoing port): Serial2/0

Holdtime : 145 sec

Version :

Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M5, RELEASE SOFTWARE (fc2)

Technical Support: <http://www.cisco.com/techsupport>

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advertisement version: 2

R3# sh cdp neighbors detail

Device ID: R2

Entry address(es):

IPv6 address: 2001:23:23:23::2 (global unicast)

IPv6 address: FE80::C802:CFF:FE24:8 (link-local)

Platform: Cisco 7206VXR, Capabilities: Router

Interface: FastEthernet0/0, Port ID (outgoing port): FastEthernet0/0

Holdtime : 160 sec

Version :

Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M5, RELEASE SOFTWARE (fc2)

Technical Support: <http://www.cisco.com/techsupport>

Copyright (c) 1986-2013 by Cisco Systems, Inc.

Compiled Fri 13-Sep-13 18:47 by prod_rel_team

advertisement version: 2

Duplex: full

R3#

Important Commands:

sh run
sh start
sh ipv6 int brief
sh ipv6 int brief | exclude unassigned
sh ipv6 int brief | exclude unassigned|down
sh int description | exclude down
sh run | section interface
sh ipv6 interface fa0/0
sh cdp neighbors detail

Ratan